

Euclid’s Elements — Book XI

Proposition 22

Formal Reconstruction — Technical Documentation

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For three plane angles, if any two together exceed the third and the containing sides are equal, the three joining chords can form a triangle.

1 Purpose and Scope

Proposition XI.22 is a striking boundary case in Book XI. It occurs in the solid-geometry book and prepares the construction of a solid angle in the next proposition, but its own mathematical content is completely planar.

The proposition begins with three proper plane angles. Each angle is contained by two segments of one common length. If the sum of any two angles is greater than the third, Euclid joins the endpoints of the equal radial segments and proves that the three resulting chords satisfy the triangle inequalities. Proposition I.22 can then construct a triangle with those three chord lengths.

The formal reconstruction preserves that structure, but makes two points explicit.

First, the project does not introduce a numerical angle measure or a general addition operation on angles. The phrase “the sum of two angles is greater than a third angle” is represented by the neutral synthetic relation `HilbertTwoAnglesGreaterThanAngle`.

Second, the final appeal to I.22 is not logically free. In the present Hilbert development, constructing a triangle from three side lengths ultimately uses circle–circle intersection. The axiom audit for XI.22 therefore exposes `hilbert_circle_circle_intersection`, even though every preceding angle-to-chord comparison is neutral and dimension-free.

Two public production entry points are retained:

```
Geometry.euclid_proposition_11_22
Geometry.euclid_proposition_11_22_wyler
```

They do not call one another. Both are proved from the same neutral planar core, and neither requires a spatial incidence class.

2 Euclid's Statement

Let the three plane angles be

$$\angle AOB, \quad \angle CPD, \quad \angle EQF,$$

and assume that the six radial segments have one common length:

$$OA = OB = PC = PD = QE = QF.$$

Assume also the three strict angle inequalities

$$\begin{aligned} \angle AOB + \angle CPD &> \angle EQF, \\ \angle CPD + \angle EQF &> \angle AOB, \\ \angle EQF + \angle AOB &> \angle CPD. \end{aligned}$$

Joining the outer endpoints gives the three chords

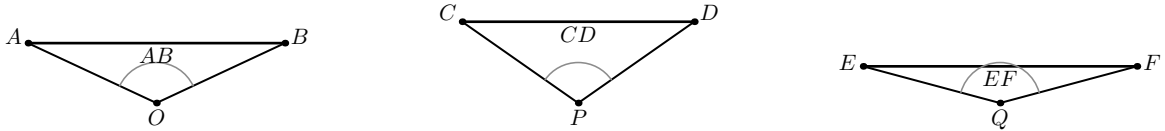
$$AB, \quad CD, \quad EF.$$

The proposition asserts that these three segments satisfy the triangle inequalities and hence can be used as the three sides of a triangle.

The historical statement is therefore not intrinsically spatial. Its role in Book XI is preparatory: XI.23 uses the triangle supplied by XI.22 in the construction of a solid angle.

3 Classical Configuration

Figure 1 displays the configuration used throughout the formal reconstruction. The three angle vertices are unrelated points in a plane-level theory; no common ambient plane needs to be named because the theorem is formulated only through point, order, congruence, and angle relations.



$$OA = OB = PC = PD = QE = QF$$

$$\angle AOB + \angle CPD > \angle EQF$$

$$\angle CPD + \angle EQF > \angle AOB$$

$$\angle EQF + \angle AOB > \angle CPD$$

Figure 1: Three equal-radius angle configurations for Proposition XI.22. The radial segments have one common length. The chords AB , CD , and EF are the three candidate side lengths.

The geometric mechanism is the familiar equal-radius chord principle: for two isosceles triangles with corresponding radial sides equal, equality of included angles gives equality of the opposite chords, while strict inequality of the included angles gives strict inequality of the opposite chords.

In the formal library these two facts are isolated as

```
hilbert_equalRadiusChord_congruent
hilbert_equalRadiusChord_less
```

in HilbertEqualRadiusChord.lean.

4 Euclid's Classical Proof

Euclid proves one triangle inequality and obtains the other two cyclically. Using the historical lettering, suppose the three angles are

$$\angle ABC, \quad \angle DEF, \quad \angle GHK,$$

with

$$AB = BC = DE = EF = GH = HK.$$

The corresponding chords are

$$AC, \quad DF, \quad GK.$$

To prove, for example,

$$AC + GK > DF,$$

Euclid constructs at H a ray HL such that

$$\angle KHL = \angle ABC$$

and lays off

$$HL = AB.$$

By SAS,

$$KL = AC.$$

The angle hypothesis

$$\angle ABC + \angle GHK > \angle DEF$$

and the placement of HL give

$$\angle GHL > \angle DEF.$$

The equal-radius comparison then yields

$$GL > DF.$$

Finally the triangle inequality in triangle GKL gives

$$GK + KL > GL,$$

hence

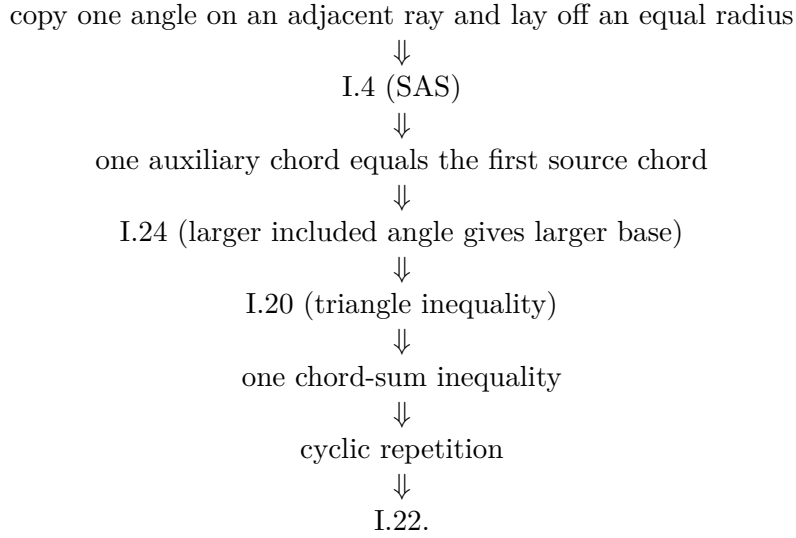
$$GK + AC > DF.$$

Cycling the construction gives

$$AC + DF > GK, \quad DF + GK > AC.$$

The classical proof then invokes I.22 to construct a triangle from AC, DF, GK .

The local classical dependency spine is



5 Why Ordinary Angle Addition Is Not Used

The expression

$$\alpha + \beta > \gamma$$

is potentially misleading if ordinary angles are treated only as the smaller angles between two rays. A numerical sum can pass through two right angles, while the project angle object remains an ordinary proper angle.

For this reason XI.20 introduced a witness-based relation rather than a numerical operation:

```

def HilbertTwoAnglesGreaterThanAngle
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A O B C P D E Q F : Geo.Point) : Prop :=
  Not (PrimCollinear Geo A O B) /\
  Not (PrimCollinear Geo C P D) /\
  Not (PrimCollinear Geo E Q F) /\
  (
    HilbertAngleLess Geo E Q F A O B
    \/
    Geo.AngleCongruent E Q F A O B
    \/
    exists X,
      HilbertRayMeetsSegment Geo Q X E F /\
      Geo.AngleCongruent A O B E Q X /\
      HilbertAngleLess Geo X Q F C P D
  )

```

The three cases have a direct geometric meaning.

1. The target angle is already smaller than the first source angle.
2. The target angle equals the first source angle.

3. The target angle decomposes into a copy of the first source angle plus a remainder smaller than the second source angle.

This relation was moved to the neutral module `HilbertAngleSumComparison.lean`. It is shared by XI.20, XI.21, and XI.22 and contains no spatial hypothesis.

6 The Equal-Radius Chord Bridge

The central reusable theorem of the XI.22 development is

```
theorem
  hilbert_twoAnglesGreaterThanAngle_equalRadius_chordSumGreater
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    ...
    (hSum :
      HilbertTwoAnglesGreaterThanAngle
        Geo A O B C P D E Q F)
    ...
    :
      HilbertSegmentSumGreater
        Geo A B C D E F
```

Its mathematical content is

$$\angle AOB + \angle CPD > \angle EQF \implies AB + CD > EF,$$

under the common-radius hypotheses.

This theorem is the genuine neutral core of XI.22. Once it is available, the public proposition is only a cyclic threefold application followed by I.22.

The proof is split according to the three constructors of `HilbertTwoAnglesGreaterThanAngle`.

6.1 First-angle-greater branch

If

$$\angle EQF < \angle AOB,$$

then equal-radius chord comparison gives

$$EF < AB.$$

Since CD is non-null,

$$EF < AB < AB + CD.$$

The corresponding formal lemma is

```
hilbert_equalRadiusChordSum_of_first_greater
```

6.2 First-angle-equal branch

If

$$\angle EQF \cong \angle AOB,$$

then SAS gives

$$EF = AB.$$

Again the non-null segment CD makes the sum strictly larger:

$$EF = AB < AB + CD.$$

The formal lemma is

hilbert_equalRadiusChordSum_of_first_equal

6.3 Decomposition branch

The third branch is the formal counterpart of Euclid's auxiliary construction. A ray QX meets the open segment EF , the angle EQX is congruent to the first source angle, and the remainder XQF is smaller than the second source angle.

A point R is laid off on the same ray with

$$QR = OA.$$

Then

$$AB = ER$$

by equal-radius SAS and

$$RF < CD$$

by the equal-radius strict chord comparison.

Figure 2 shows this branch.

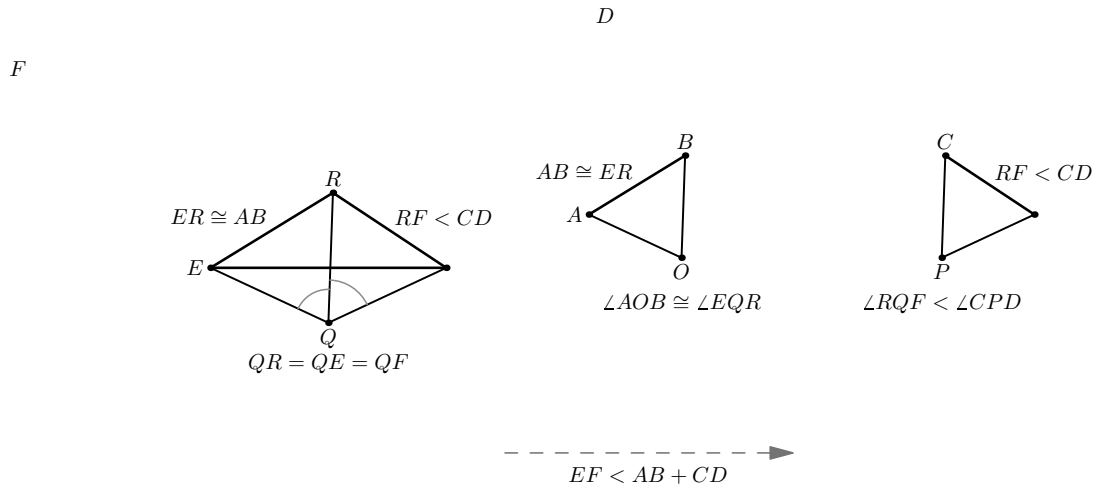


Figure 2: The decomposition branch. The ray QR carries a copy of the first source angle and the common radius. Thus $ER \cong AB$, while the remainder comparison gives $RF < CD$. The proof then converts these two facts into $EF < AB + CD$.

The formal proof deliberately handles both possibilities for E, R, F .

If they are noncollinear, I.20 applied to triangle ERF gives a segment representing $ER + RF$ and strictly exceeding EF . Replacing ER by AB and RF by the longer segment CD yields

$$EF < AB + CD.$$

If they are collinear, the interior-ray crossing forces the laid-off point R to coincide with the crossing point on the open segment EF . Hence

$$E - R - F,$$

so

$$EF = ER + RF < AB + CD.$$

This second branch avoids introducing a separate theorem saying that a line meets an equal-radius circle in at most two points.

The proposition-local lemmas recording this split are

```
hilbert_equalRadiusChordSum_of_decomposition_core
hilbert_chordSum_of_collinear_decomposition
hilbert_equalRadiusChordSum_of_decomposition
```

7 Synthetic Segment Addition

The project does not use numerical segment lengths. A strict inequality

$$AB + CD > EF$$

is represented by `HilbertSegmentSumGreater`: one lays off a copy of CD beyond B , obtaining a representative segment for the sum, and then compares EF with that representative.

The neutral helper module `HilbertSegmentSum.lean` isolates the corresponding operations:

```
hilbert_segmentSum_witness
hilbertSegmentSumGreater_of_less_first
hilbertSegmentSumGreater_of_congruent_first
hilbertSegmentSumGreater_swap
```

This layer is important for XI.22 because the final I.22 interface expects exactly the three strict segment-sum inequalities, not a numerical length equation.

8 The Formal Proof DAG

After the neutral chord theorem has been proved, XI.22 has a very small public proof graph:

$$\begin{array}{ccc} \angle AOB + \angle CPD > \angle EQF & \angle CPD + \angle EQF > \angle AOB & \angle EQF + \angle AOB > \angle CPD \\ \Downarrow & \Downarrow & \Downarrow \\ AB + CD > EF & CD + EF > AB & EF + AB > CD \end{array}$$

followed by

$$\begin{array}{c}
 AB + CD > EF, \quad CD + EF > AB, \quad EF + AB > CD \\
 \Downarrow \\
 \text{I.22} \\
 \Downarrow \\
 \text{a triangle with side lengths } AB, CD, EF.
 \end{array}$$

There is no spatial node in this graph.

9 Direct Hilbert Reconstruction

The production theorem is

```

theorem euclid_proposition_11_22
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A O B C P D E Q F : Geo.Point)
  (h12_3 :
    HilbertTwoAnglesGreaterThanAngle
      Geo A O B C P D E Q F)
  (h23_1 :
    HilbertTwoAnglesGreaterThanAngle
      Geo C P D E Q F A O B)
  (h31_2 :
    HilbertTwoAnglesGreaterThanAngle
      Geo E Q F A O B C P D)
  ...
  :
  exists K : Geo.Point,
    Geo.Congruent C K A B /\
    Geo.Congruent D K E F

```

The third side of the constructed triangle is CD , while the two new sides are congruent to AB and EF . This is exactly the form supplied by the existing production theorem `euclid_proposition_22`.

The proof has four stages:

1. extract properness of the three angles and hence non-nullity of the three chords;
2. apply the neutral angle-to-chord-sum theorem to the first cyclic inequality;
3. rebase the common-radius congruences twice and apply the same neutral theorem to the remaining two cyclic inequalities;
4. normalize the order of one segment sum and call I.22.

No earlier proposition of Book XI is called by the production theorem.

10 Hilbert–Wyler Reconstruction

The independent Wyler entry point is


```

theorem euclid_proposition_11_22_wyler
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  ...
  :
  exists K : Geo.Point,
    Geo.Congruent C K A B /\
    Geo.Congruent D K E F

```

Its production file is

```
CGJteamLab/Wyler/Proposition11_22.lean
```

This file does not import or call `euclid_proposition_11_22`. It independently performs the three cyclic applications of the common neutral chord theorem and then calls I.22.

Unlike XI.20 and XI.21, there is no point at which the Hilbert and Wyler routes need different spatial infrastructure. In particular XI.22 requires neither

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertWylerAxioms

```

nor any local flat, exchange, or dimension assumption.

Thus the two public entry points differ only as architectural endpoints. Their mathematical proof DAG is identical.

11 Planar and Dimension-Free Status

XI.22 is therefore stronger architecturally than its location in Book XI might suggest.

1. The statement mentions only points, proper angles, congruent segments, synthetic angle comparison, and segment-sum comparison.
2. The equal-radius chord lemmas are ordinary planar SAS/I.24 facts.
3. The decomposition branch uses only order, angle comparison, congruence, and I.20.
4. The final construction is I.22.
5. No proposition-specific spatial incidence statement occurs.

Consequently the same theorem can be used unchanged inside E^3 , E^4 , or a general E^n development whenever the common planar Hilbert metric layer is available. No local 3-flat reduction is required.

This is an instructive contrast with XI.20 and XI.21. Those propositions are dimension-free after a small ambient incidence obligation has been discharged. XI.22 has no such obligation at all.

12 Continuity Boundary and Proposition I.22

The final call is

```
euclid_proposition_22
```

with the three chords AB, CD, EF as the prescribed side lengths.

The existing I.22 theorem takes the three strict segment-sum inequalities and constructs a point K satisfying

$$CK \cong AB, \quad DK \cong EF,$$

with CD as the third side.

This is precisely where circle continuity enters. The angle-to-chord reduction itself is neutral; the existence of the final triangle is not. The formal dependency therefore clarifies a point hidden in the classical presentation: proving the three triangle inequalities and constructing the triangle from them are distinct logical steps.

The historical proof simply cites I.22 at this point. The formal theorem makes the construction boundary explicit through the already audited implementation of I.22.

13 Axiom Audit

The audit commands are

```
#print axioms Geometry.euclid_proposition_11_22
#print axioms Geometry.euclid_proposition_11_22_wyler
```

and Lean reports the same dependency set for both:

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Geometry.bookZero_nullSegment2,
 Geometry.hilbert_circle_circle_intersection,
 Quot.sound]
```

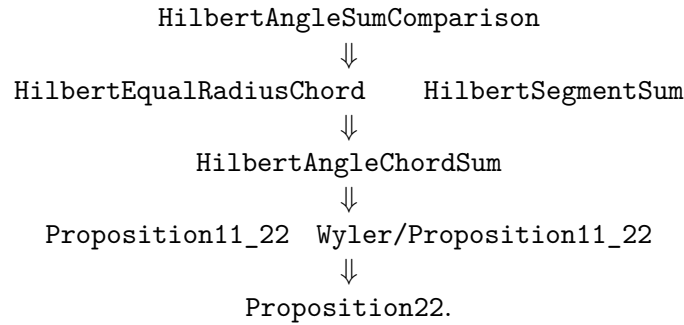
The equality of the two audit results confirms that the Wyler entry point adds no Wyler-specific axiom and that the direct entry point adds no spatial Hilbert axiom.

The occurrence of `hilbert_circle_circle_intersection` comes from the final I.22 construction. It should not be attributed to the angle comparison or equal-radius chord machinery.

The two `bookZero_nullSegment` declarations are inherited elementary segment infrastructure. No new XI.22-specific axiom constant is introduced.

14 Dependency Summary

The production architecture is



The status can be summarized as follows:

New proposition-specific geometric axiom	no,
Numerical angle measure	no,
General angle addition operation	no,
Spatial incidence	no,
HilbertSpacePrimitive	no,
HilbertWylerAxioms	no,
Wyler I.7 / Smith I.5 / exchange	no,
Ambient dimension bound	no,
Coordinates / vectors / scalar product	no,
Circle continuity	yes, through I.22,
sorry/admit	no,
Full repository build	clean.

15 Conclusion

Proposition XI.22 is formally valuable for a reason that is easy to miss in the classical organization of the *Elements*. Although it belongs to Book XI and prepares a solid-angle construction, the proposition itself is a pure theorem of planar metric geometry.

The main formal work is not spatial. It is the conversion

$$\text{two-angle comparison} \implies \text{two-chord sum comparison.}$$

The witness-based relation `HilbertTwoAnglesGreaterThanAngle` makes this conversion possible without numerical angle measure. Equal-radius SAS and I.24 handle the simple branches, while the decomposition branch reconstructs Euclid’s auxiliary ray synthetically and covers both the collinear and noncollinear placements of the laid-off point.

After that neutral theorem is isolated, XI.22 itself collapses to three cyclic applications and one invocation of I.22. The identical Hilbert and Hilbert–Wyler axiom audits show that there is no residual spatial content.

The only non-neutral boundary is constructive: I.22 uses circle–circle intersection. Thus XI.22 provides a particularly clean example of the architectural distinction between neutral comparison geometry and continuity needed for existence constructions.